

HHI Metadata Protection Plan

Enterprise Governance Standard for Metadata Integrity, Provenance Binding,

Telemetry Continuity, and Stop Authority Protection

Governing all metadata across the HHI ecosystem — with zero-trust enforcement,

immutable audit chains, and cryptographic accountability at every tier.

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INTERNAL – HHI GOVERNANCE RESTRICTED

Table of Contents

1. Executive Summary

1.1 Purpose and Scope

1.2 Foundational Importance of Metadata Integrity

1.3 Protection Pillars

1.4 Statement of Authority

2. Scope and Applicability

2.1 In-Scope Systems and Assets

2.2 In-Scope Metadata Types

2.3 Exclusions and Boundary Conditions

2.4 Relationship to Upstream HHI Governance Policies

3. Definitions and Terminology

4. Metadata Integrity Framework

4.1 Integrity Principles

4.2 Metadata Classification Tiers

4.3 Integrity Enforcement Points

5. Checksum Anchoring

5.1 Purpose and Scope

5.2 Checksum Generation Protocol

5.3 Anchor Verification Procedures

5.4 Anchor Rotation and Key Management

6. Provenance Binding

6.1 Provenance Architecture

6.2 Provenance Record Structure

6.3 Provenance Binding Enforcement

6.4 Cross-Entity Provenance Tracing

7. Telemetry Continuity Enforcement

7.1 Continuity Definition and Objectives

7.2 Continuity Enforcement Architecture

7.3 Continuity Violation Response

7.4 Telemetry Retention and Archival

8. Stop Authority Protection

8.1 Stop Authority Definition

8.2 Stop Authority Metadata Protections

8.3 Stop Authority Invocation Integrity

8.4 Stop Authority Chain of Custody

9. Cross-Repo Corroboration

9.1 Corroboration Architecture

9.2 Corroboration Protocol

9.3 Discrepancy Classification and Remediation

9.4 Corroboration Reporting

10. Immutable Audit Chains

10.1 Audit Chain Architecture

10.2 Audit Record Schema

10.3 Audit Chain Integrity Verification

10.4 Audit Chain Storage and Replication

11. Runtime Drift Detection

11.1 Drift Definition and Taxonomy

11.2 Drift Detection Engine

11.3 Drift Scoring and Severity

11.4 Drift Response Procedures

11.5 Drift Detection Tooling Requirements

12. Governance Controls and Roles

12.1 Roles and Responsibilities (RACI)

12.2 Access Control Requirements

12.3 Change Management

13. Compliance, Audit, and Review

13.1 Internal Audit Requirements

13.2 External Audit Provisions

13.3 Policy Review Cycle

13.4 Key Performance Indicators

14. Appendices

Appendix A – Checksum Algorithm Reference

Appendix B – Provenance Record Templates

Appendix C – Drift Detection Configuration Reference

Appendix D – Corroboration Quorum Rules

Appendix E – Document Control and Revision History

1. Executive Summary

1.1 Purpose and Scope

The **HHI Metadata Protection Plan** (hereinafter "this Plan" or "MDP") establishes binding governance standards, technical controls, enforcement protocols, and accountability structures for the protection of all metadata assets within the Human-Human Interface (HHI) ecosystem.

This Plan defines the mandatory requirements that govern how metadata is created, bound, stored, verified, corroborated, and audited across every HHI repository, runtime environment, telemetry pipeline, and governance artifact.

This Plan is not advisory. It is an enforceable governance standard with the same authority as any other instrument in the HHI Governance Library. All HHI system participants — whether human operators, automated agents, integration pipelines, or third-party service connectors — are bound by the provisions of this Plan upon its effective date.

The scope of this Plan spans the full metadata lifecycle: from initial write-time attestation through archival sealing, from individual field-level integrity to cross-repository corroboration, and from routine telemetry continuity monitoring to the highest-precedence Stop Authority protections.

1.2 Foundational Importance of Metadata Integrity

Within the HHI ecosystem, metadata is not supplementary information. It is the structural substrate upon which governance decisions, trust determinations, provenance chains, audit trails, and operational signals are constructed. Compromised, drifted, or unverified metadata does not merely create data quality problems — it fundamentally undermines the integrity of every decision, action, and record that depends on it.

The HHI governance doctrine treats metadata integrity as a *first-order security property*, equivalent in criticality to access control or cryptographic confidentiality. An adversary or fault condition that can alter metadata without detection can, in principle, falsify the provenance of any record, suppress Stop Authority signals, sever audit chain continuity, or manufacture a misleading governance history.

This Plan operationalizes that doctrine by imposing cryptographic, procedural, and architectural controls that collectively ensure metadata cannot be silently corrupted, forged, or erased — and that any attempt to do so is detected, logged, and escalated without exception.

1.3 Protection Pillars

This Plan is organized around seven interlocking protection pillars, each addressing a distinct dimension of metadata integrity risk:

1. **Checksum Anchoring** — Cryptographic hash anchoring of all Tier 0 and Tier 1 metadata records to tamper-evident reference points.
2. **Provenance Binding** — End-to-end provenance record attachment tracing every metadata record to its generating entity, activity, and time context.
3. **Telemetry Continuity Enforcement** — Sequence-numbered, gap-detected, heartbeat-verified telemetry streams that treat any discontinuity as an integrity violation.
4. **Stop Authority Protection** — Maximum-protection regime for the highest-precedence governance control signals in the HHI ecosystem.
5. **Cross-Repo Corroboration** — Multi-repository consensus validation ensuring no single-point manipulation of canonical metadata records.

6. **Immutable Audit Chains** — Hash-linked, append-only audit records capturing every metadata lifecycle event with cryptographic continuity.
7. **Runtime Drift Detection** — Continuous baseline comparison identifying unauthorized deviations in metadata values, schemas, relationships, and temporal markers.

1.4 Statement of Authority

▲ STATEMENT OF AUTHORITY

This Plan governs **all metadata** across the HHI ecosystem without exception. No individual system, team, automated agent, or external party may assert exemption from any provision of this Plan without a documented waiver approved by the HHI Governance Architects Council and recorded in the official HHI Governance Registry. Any waiver must include a compensating control attestation, a time-bounded scope, and a defined remediation path. Unapproved deviations from this Plan constitute a governance violation subject to escalation, remediation, and audit action.

2. Scope and Applicability

2.1 In-Scope Systems and Assets

This Plan applies to all systems, environments, and infrastructure components that form part of or interface with the HHI ecosystem, including but not limited to:

- All HHI primary, secondary, and federated **repositories**, whether hosted centrally or at distributed nodes
- All **runtime environments** executing HHI governance logic, agents, or data pipelines
- All **telemetry pipelines** transmitting operational signals, health data, or governance events
- All **governance artifacts** including policy documents, schema registries, epoch markers, and governance epoch roots
- All **integration connectors** linking HHI systems to external data sources, partner systems, or third-party services
- All **archival and cold storage** systems holding HHI metadata records
- All **audit infrastructure** including log aggregators, audit chain repositories, and monitoring platforms

2.2 In-Scope Metadata Types

The following categories of metadata are explicitly governed by this Plan:

- **Structured metadata** — Schema-defined fields, typed records, and structured annotations attached to HHI data assets
- **Provenance records** — Origin, transformation, and consumption records associated with any HHI data asset
- **Audit logs** — All records of system events, access events, modification events, and governance decisions
- **Identity bindings** — Records associating metadata or data assets with a specific user, agent, or system identity
- **Schema registries** — Canonical definitions of metadata field types, required fields, validation rules, and versioning records
- **Telemetry continuity records** — Sequence numbers, gap reports, heartbeat records, and continuity attestations
- **Stop Authority records** — All records bearing Stop Authority classification, invocation signals, or chain-of-custody attestations
- **Governance epoch markers** — Records defining the boundaries, roots, and authorities of HHI governance epochs

2.3 Exclusions and Boundary Conditions

The following are explicitly excluded from the scope of this Plan, unless they are determined to intersect with an in-scope metadata type:

- Raw data payloads that contain no metadata fields (governed by HHI Data Classification Policy)
- Transient in-memory metadata that is never persisted to a repository or transmitted to an external system
- Development and sandbox environments designated as non-production and explicitly excluded in the HHI Environment Registry — though such environments must not process real Tier 0 or Tier 1 metadata
- Third-party system metadata that never enters an HHI repository or is processed by HHI governance logic

i Boundary Condition Note

When ambiguity exists regarding whether a metadata type or system falls within scope, the determination defaults to **in-scope** pending a formal boundary assessment by the Metadata Integrity Officer. The burden of proof for exclusion rests with the requesting party.

2.4 Relationship to Upstream HHI Governance Policies

This Plan operates within the broader HHI Governance Framework and is subordinate only to the HHI Master Governance Charter. It is a peer instrument to the HHI Data Classification Policy, HHI Identity and Access Management Standard, and HHI Incident Response Plan. Where conflicts arise between this Plan and peer policies, the provision imposing the *stricter*

control shall prevail, pending resolution by the HHI Governance Architects Council. This Plan takes precedence over any system-level or team-level metadata handling procedure.

3. Definitions and Terminology

The following terms carry precise, binding meanings throughout this Plan. Where a term defined below conflicts with a colloquial or industry-general meaning, the definition below shall govern.

Term	Definition
Metadata Integrity	The property of a metadata record whereby its content, structure, provenance, and relationships are demonstrably unaltered from their last verified canonical state, as confirmed by cryptographic and procedural controls.
Provenance Binding	The act of attaching a cryptographically signed, standards-compliant provenance record to a metadata artifact at write time, establishing an irrefutable link between the artifact and its generating entity, activity, and temporal context.
Checksum Anchoring	The process of computing a cryptographic hash of a metadata record and registering that hash against an immutable reference point (governance epoch root, notarized timestamp, or distributed ledger anchor), such that any subsequent modification of the record is detectable.
Telemetry Continuity	The property of a telemetry stream whereby every expected event is received in correct sequence, no events are silently dropped or reordered without detection, and any gap or override is immediately classified as an integrity violation.

Stop Authority	The highest-precedence governance control signal in the HHI ecosystem, capable of halting, suspending, or constraining any automated process, agent, or system action. Stop Authority records receive the maximum protection regime defined in this Plan.
Drift Detection	The automated, continuous process of comparing live metadata values, schemas, and relationships against a verified canonical baseline, and identifying any deviation not attested through the approved change management process.
Cross-Repo Corroboration	The process by which the integrity of a metadata record in one HHI repository is validated by comparison against independent copies or derivations of the same record in at least two other HHI repositories, according to a defined quorum rule.
Immutable Audit Chain	A hash-linked, append-only sequence of audit records, modeled on Merkle tree principles, in which each record includes the hash of the preceding record, making any retroactive alteration of the chain computationally detectable.
Schema Registry	The authoritative, version-controlled repository of canonical metadata field definitions, types, constraints, and validation rules for the HHI ecosystem. The Schema Registry is itself governed as a Tier 1 metadata asset.
Governance Epoch	A defined period of HHI governance activity bounded by a start marker and an end marker, each of which is a Tier 0 metadata record. Governance epochs provide the temporal frame

	of reference for provenance records, audit chains, and corroboration operations.
Canonical Record	The single authoritative version of a metadata record, as determined by the Schema Registry, provenance binding, and corroboration quorum. Where multiple copies of a record exist, the canonical record is the one designated as authoritative following corroboration resolution.
Attestation Token	A digitally signed, time-stamped token issued by an authorized HHI governance actor or system, asserting that a specific metadata record, operation, or state has been reviewed and approved under the applicable governance controls.
Quorum Validation	A corroboration or authorization outcome reached when a defined minimum number of independent HHI repositories or authorized actors agree on the validity, content, or status of a metadata record or governance decision, per the quorum rules defined in Appendix D.

4. Metadata Integrity Framework

4.1 Integrity Principles

The HHI Metadata Integrity Framework is founded on four inviolable principles that govern every control, protocol, and architectural decision in this Plan:

1. **Immutability:** Once a metadata record has been written and attested, its content cannot be altered. Modifications are represented as new versioned records linked to the original via the audit chain and provenance binding. The original record is preserved in its sealed state.
2. **Non-Repudiation:** Every metadata write, modification, read with consequence, and governance decision must be attributable to a specific, authenticated actor (human or system). Attestation tokens and audit records must make it impossible for any actor to credibly deny their role in a metadata event.

3. **Auditability:** The complete lifecycle of every metadata record — from creation through archival — must be reconstructible from the immutable audit chain and provenance records without gaps, ambiguities, or dependency on the goodwill of any individual actor.
4. **Traceability:** Every metadata record must be traceable to its origin through an unbroken provenance chain. Derived or transformed records must carry explicit derivation bindings referencing their source records.

⚠️ ZERO-TRUST METADATA POSTURE

The HHI ecosystem operates under a **zero-trust metadata posture**: every metadata field, record, and relationship is treated as **untrusted until cryptographically verified** against its registered checksum and validated provenance binding. No metadata value is accepted as authoritative based solely on its source system, channel, or claimed identity. Verification is mandatory at every enforcement point defined in Section 4.3.

4.2 Metadata Classification Tiers

All HHI metadata is classified into one of four protection tiers at the time of schema registration. Tier classification determines the applicable protection controls, verification cadence, retention requirements, and escalation thresholds defined throughout this Plan.

Tier	Classification	Examples	Required Protections
Tier 0	Critical Governance	Stop Authority records, governance epoch markers, provenance roots, canonical record designations, anchor key registrations	Checksum anchoring (SHA-3-512); dual-custody attestation; write-once storage; real-time cross-repo corroboration (≥ 3 nodes); immutable audit chain; drift detection ≤ 30 s; Stop Authority notification on any anomaly
Tier 1	Operational Integrity	Checksums, hash chains, schema versions, identity bindings, attestation tokens, corroboration	Checksum anchoring (SHA-256 or SHA-3-512); single-custody attestation with CAB

		reports, schema registry records	approval for changes; corroboration ≥ 2 nodes; audit chain; drift detection ≤ 5 min
Tier 2	Telemetry Continuity	Runtime logs, drift detection records, corroboration signals, sequence numbers, heartbeat records, gap reports	Checksum at write time; single-custody attestation; corroboration ≥ 2 nodes; audit chain; drift detection ≤ 15 min; retention per Section 7.4
Tier 3	Reference Metadata	Labels, tags, annotations, non-binding descriptors, human-readable comments, display metadata	Integrity hash at write time; single-node corroboration acceptable; audit chain (read events not required); drift detection ≤ 1 hour

Tier Escalation Principle

If a Tier 3 or Tier 2 metadata record is discovered to influence a governance decision, Stop Authority signal, or provenance chain, it must be **immediately re-classified** to the appropriate higher tier and retrospectively subject to that tier's protection requirements, including checksum anchoring and audit chain backfill where technically feasible.

4.3 Integrity Enforcement Points

Metadata integrity controls are applied at four mandatory enforcement points in the metadata lifecycle. No metadata record may bypass any enforcement point applicable to its tier.

- **Ingest-Time Validation:** All incoming metadata is validated against its registered schema, checked for required fields, and assigned a provisional tier classification before being written to any HHI repository. Invalid or unclassifiable records are quarantined pending review.
- **Write-Time Attestation:** At the moment of persistence, a cryptographic hash of the canonical serialized record is computed, an attestation token is generated by the writing entity, and the hash-attestation pair is registered in the checksum anchor registry. Provenance binding is attached simultaneously.
- **Read-Time Verification:** For Tier 0 and Tier 1 records, every read operation that will inform a governance decision or trigger a system action must include a verification step comparing

the record's current hash against the registered anchor. A failed verification halts the operation and triggers an alert.

- **Archive-Time Sealing:** At the end of a retention period or upon explicit archival, a Tier 0 or Tier 1 record is cryptographically sealed with a timestamped, notarized hash. The seal is registered in the audit chain and the archival attestation token is stored alongside the sealed record.

5. Checksum Anchoring

5.1 Purpose and Scope

Checksum anchoring provides the primary cryptographic assurance that a metadata record has not been altered since its last verified state. By binding the hash of a record to an external, immutable reference point — a governance epoch root, a distributed ledger anchor, or a notarized timestamp — checksum anchoring makes tampering detectable even if the record's storage medium is compromised.

Checksum anchoring is **mandatory for all Tier 0 and Tier 1 metadata records**. Tier 2 records receive write-time hashing. Tier 3 records receive a best-effort integrity hash. The approved algorithms are SHA-256 (for Tier 1) and SHA-3-512 (mandatory for Tier 0 and recommended for Tier 1 where computational resources permit). Deprecated algorithms are enumerated in Appendix A.

5.2 Checksum Generation Protocol

The following protocol governs checksum generation for all in-scope metadata records:

1. **Canonicalize the record** using JSON Canonical Form as defined in RFC 8785. All fields must be serialized in the canonical order defined by the HHI Schema Registry for the record type. Whitespace normalization and Unicode normalization (NFC) are mandatory.
2. **Compute the hash** of the canonicalized byte representation using the approved algorithm for the record's tier. For Tier 0: SHA-3-512. For Tier 1: SHA-256 minimum, SHA-3-512 preferred.
3. **Include in the hash computation** the following mandatory fields (in canonical order): `recordID, schemaVersion, tier, epochID, actorID, timestamp (ISO 8601 UTC), recordBody, provenanceToken`. No field listed here may be excluded.
4. **Register the computed hash** in the HHI Checksum Anchor Registry, binding it to: the current governance epoch root hash, a notarized UTC timestamp, and the attestation token of the registering actor.
5. **Store the checksum registration record** in the immutable audit chain (Section 10) and the corroboration-eligible checksum store.

5.3 Anchor Verification Procedures

On-Demand Verification: Any authorized actor or system may request verification of a specific record's checksum at any time. The verification process recomputes the hash of the current record state using the canonical protocol above and compares it against the registered anchor. The result — match or mismatch — is recorded in the audit chain regardless of outcome.

Scheduled Verification Cadence:

Tier	Scheduled Verification Frequency	Verification Method
Tier 0	Continuous (every 30 seconds)	Automated re-hash against epoch root anchor
Tier 1	Every 5 minutes	Automated re-hash against registered checksum
Tier 2	Every 15 minutes	Automated spot verification (sampled)
Tier 3	Hourly (sampled, 10% of records)	Automated spot verification

Mismatch Handling: A checksum mismatch triggers the following escalation path: (1) Immediate automated quarantine of the affected record; (2) Alert routed to the Metadata Integrity Officer and the relevant Repository Custodian; (3) Drift Detection Engine notified to initiate forensic baseline comparison; (4) If Tier 0, immediate Stop Authority notification; (5) Incident record created in the audit chain; (6) No reads of the quarantined record for governance purposes until resolution is attested.

5.4 Anchor Rotation and Key Management

Checksum anchor keys — the cryptographic signing keys used to register and authenticate anchor registrations — are governed by the following lifecycle rules:

- **Creation:** Anchor keys are generated in a hardware security module (HSM) or equivalent isolated cryptographic environment. Key generation events are logged in the audit chain with full attestation.

- **Rotation:** Tier 0 anchor keys must be rotated at least every 90 days or immediately following any suspected compromise. Tier 1 anchor keys rotate at least every 180 days. Rotation events are corroborated across at least two repositories.
 - **Revocation:** Compromised or expired anchor keys are revoked immediately, with revocation records written to the immutable audit chain and propagated to all corroboration nodes within 15 minutes.
 - **Dual-Custody for Tier 0:** Tier 0 anchor key operations (creation, rotation, revocation) require the authenticated participation of at least two authorized Stop Authority Custodians, acting independently. No single actor may perform a Tier 0 anchor key operation unilaterally.
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6. Provenance Binding

6.1 Provenance Architecture

Provenance binding ensures that every metadata record in the HHI ecosystem carries a verifiable record of its origin, the activities that produced or transformed it, and the agents responsible for those activities. The HHI provenance model follows the W3C PROV Data Model (PROV-DM) as its foundational standard, extended with HHI-specific fields to capture governance epoch context and attestation token linkage.

The provenance chain for any HHI metadata record spans three stages:

1. **Origin:** The entity from which the record derives its existence — a system event, a governance decision, a data ingestion, or a human action.
2. **Transformation:** Any activity that modified, derived, aggregated, or versioned the record from its origin state. Each transformation is a separately bound provenance record.
3. **Consumption:** Any use of the record that produces a governance consequence — a decision, an action, a downstream derivation. Consumption events are logged in the audit chain but may optionally carry a provenance binding when they produce new artifacts.

6.2 Provenance Record Structure

All provenance records attached to HHI metadata must conform to the following schema. Fields marked **Required** must be present for the record to be considered validly bound.

Field	Type	Required	Description
<code>prov:entity</code>	URI	Required	Unique identifier (URI) of the metadata

			artifact to which this provenance record is bound.
prov:activity	URI	Required	Identifier of the activity (write, transform, version, governance decision) that produced or modified the entity.
prov:agent	URI	Required	Authenticated identifier of the entity (user, system, automation agent) responsible for the activity.
prov:wasGeneratedBy	URI	Required	Reference to the activity that generated the entity. Links entity to its producing activity.
prov:wasDerivedFrom	URI	Conditional	Reference to the source entity from which this record was derived. Required for all transformed or versioned records.
prov:startedAtTime	ISO 8601 UTC	Required	Timestamp at which the generating activity commenced.
prov:endedAtTime	ISO 8601 UTC	Required	Timestamp at which the generating activity

			concluded and the entity was finalized.
<code>prov:wasAssociatedWith</code>	URI	Required	The agent associated with the generating activity. Must match <code>prov:agent</code> or reference a responsible delegation chain.
<code>hhi:epochID</code>	String (UUID)	Required	The governance epoch identifier active at the time of record generation. Links the provenance record to its temporal governance frame.
<code>hhi:attestationToken</code>	JWT (signed)	Required	Digitally signed attestation token issued by the generating agent, binding the agent's authenticated identity to this provenance record at the stated time.

6.3 Provenance Binding Enforcement

Provenance binding is **mandatory at write time for all Tier 0 and Tier 1 metadata records**. Write operations that do not attach a valid, schema-conforming provenance record are rejected by the HHI repository layer before persistence. This rejection is logged in the audit chain.

Orphan Record Detection: The HHI corroboration and integrity systems continuously scan for orphan records — metadata records without a valid, resolvable provenance binding. Upon detection:

- The record is immediately quarantined and flagged as `PROVENANCE_ORPHAN`
- An alert is raised to the Metadata Integrity Officer and the relevant Repository Custodian

- A 24-hour remediation window is opened; the Custodian must either supply a valid provenance binding or escalate to the Governance Architects Council
- If no valid provenance can be established after 24 hours, the record is sealed in quarantine with an `UNRESOLVABLE_PROVENANCE` classification and preserved for audit purposes but excluded from all governance operations

Provenance Lineage Visualization: The HHI governance tooling must provide a provenance lineage visualization capability allowing authorized actors to render the complete provenance graph of any metadata record — from its origin through all transformations to its current state — in a human-readable, auditable format. Lineage visualization requests are logged in the audit chain.

6.4 Cross-Entity Provenance Tracing

HHI metadata frequently crosses repository boundaries — records originating in one repository are consumed, derived from, or corroborated in another. Provenance binding must survive these boundary crossings intact. The following requirements govern cross-entity provenance tracing:

- When a metadata record crosses a repository boundary, the receiving repository must validate the provenance binding's attestation token before accepting the record
- The receiving repository appends a **boundary crossing event** to the record's provenance chain, noting the sending repository, receiving repository, timestamp, and the validating actor's attestation token
- Federated provenance resolution — the ability to traverse a provenance chain that spans multiple HHI repositories — is a required capability of the HHI governance tooling layer
- Provenance records from decommissioned or offline repositories must be archived in the HHI Federated Provenance Archive and remain resolvable for the full retention period applicable to their tier

7. Telemetry Continuity Enforcement

7.1 Continuity Definition and Objectives

Telemetry continuity is the property of the HHI telemetry stream whereby every expected event is present, sequenced correctly, and verifiably unmodified. A continuous telemetry stream is one in which there are no gaps, no silent drops, no unauthorized overrides, and no out-of-order sequences that are not explainable and attested.

⚠ **CONTINUITY AS A SECURITY PROPERTY**

Telemetry discontinuities are not treated as operational gaps, network hiccups, or system quirks in the HHI ecosystem. They are classified as **integrity violations** and handled under the integrity violation response framework. A system that cannot maintain telemetry continuity is a system that cannot be trusted. Discontinuity creates conditions under which Stop Authority signals, governance events, and drift alerts may be suppressed, delayed, or fabricated without detection.

7.2 Continuity Enforcement Architecture

Sequence Numbering Protocol: Every telemetry event emitted by an HHI system component is assigned a monotonically increasing, globally unique sequence number within its telemetry channel. Sequence numbers are assigned by the emitting component and validated by the receiving aggregator. Gaps in sequence numbers are immediately flagged.

Gap Detection: The telemetry aggregator maintains an expected sequence number for each active channel. Upon receipt of each event, the aggregator compares the event's sequence number against the expected value. Any discrepancy — whether the received number is higher (gap) or lower (reorder) than expected — triggers the gap detection response protocol.

Heartbeat Injection: Each telemetry source injects a synthetic heartbeat event at a defined interval (default: every 10 seconds for Tier 0-relevant sources, every 60 seconds for Tier 2 sources). The heartbeat carries the current sequence number, a timestamp, and a channel health attestation. Absence of a heartbeat beyond 2× the defined interval is treated as a continuity violation.

Out-of-Order Detection and Resequencing: A resequencing buffer of defined depth (configurable per channel, default: 100 events) absorbs transient out-of-order deliveries without triggering a violation. Events that cannot be resequenced within the buffer depth or within 30 seconds are classified as a continuity violation.

7.3 Continuity Violation Response

Continuity violations are classified by severity, and automated and manual response actions are defined for each level:

Violation Class	Definition	Automated Response	Manual Escalation Trigger
Minor Gap	1–3 missing events in a non-critical channel; gap resolved within resequencing buffer window	Log to audit chain; increment gap counter; alert if threshold exceeded	If gap counter exceeds 5 within 1 hour

Major Gap	4+ missing events, or any gap in a Tier 0-relevant channel, or gap persisting beyond buffer window	Quarantine affected channel; notify Drift Detection Engine; raise integrity alert	Immediate; Metadata Integrity Officer paged
Silent Drop	Heartbeat absent beyond 2× interval with no prior shutdown attestation; suggests suppression or system failure	Channel suspended; cross-channel corroboration initiated; integrity violation recorded	Immediate; Stop Authority Custodian notified if Tier 0 channel affected
Forced Override	Evidence of deliberate sequence number manipulation, timestamp falsification, or channel substitution	Full channel shutdown; all derived records quarantined; Stop Authority notification	Immediate; Stop Authority invoked; incident declared

7.4 Telemetry Retention and Archival

Telemetry data is retained for the minimum periods specified below. Retention periods may be extended by governance decision or legal hold, but may not be shortened. At the end of the active retention period, telemetry data is sealed in an immutable archival format and transferred to write-once archival storage.

Telemetry Category	Tier	Active Retention	Archival Retention	Archival Format
Governance event telemetry	Tier 0	5 years	Indefinite	JSONL + Merkle proof
Stop Authority telemetry	Tier 0	7 years	Indefinite	JSONL + Merkle proof

Operational integrity logs	Tier 1	3 years	7 years	JSONL + SHA-256 manifest
Runtime drift records	Tier 2	2 years	5 years	JSONL + SHA-256 manifest
Reference metadata telemetry	Tier 3	1 year	3 years	Compressed JSONL

8. Stop Authority Protection

8.1 Stop Authority Definition

Stop Authority is the highest-precedence governance control signal in the HHI ecosystem. A Stop Authority signal has the capacity to halt, suspend, constrain, or override any automated process, agent, pipeline, or system action within the HHI environment, regardless of that action's normal authorization level or operational priority.

Stop Authority records — including invocation records, chain-of-custody attestations, acknowledgment records, and revocation records — are classified as **Tier 0 Critical Governance** metadata and receive the maximum protection regime defined throughout this Plan, supplemented by the additional protections specified in this section.

ABSOLUTE PROHIBITION

No automated process, agent, pipeline, or system component may alter, suppress, delay, reinterpret, reclassify, or otherwise interfere with a Stop Authority record or signal. Any system behavior that has the effect of reducing the fidelity, accessibility, timeliness, or enforceability of a Stop Authority signal — whether through design, misconfiguration, or fault — is a critical integrity violation and must be reported immediately to the Stop Authority Custodians and the Metadata Integrity Officer.

8.2 Stop Authority Metadata Protections

Stop Authority metadata is subject to all Tier 0 protections defined in this Plan, plus the following additional mandatory requirements:

- **Air-Gapped or Write-Once Storage:** All Stop Authority records must be stored on write-once media or in a logically air-gapped repository that rejects all modification operations at the infrastructure level. Read operations are logged; write operations beyond initial creation are prohibited by the storage layer itself.
- **Mandatory Dual-Custody Attestation:** Any operation on a Stop Authority record other than a read — including creation, chain-of-custody update, and any form of administrative action — requires the independent, authenticated attestation of at least two authorized Stop Authority Custodians. Dual-custody attestations are recorded in the audit chain with both actors' tokens.
- **Dedicated Signing Key:** Stop Authority records are cryptographically sealed using a dedicated Stop Authority signing key, separate from all other HHI anchor keys. This key is managed under the dual-custody protocol and rotated every 60 days.
- **Real-Time Geographic Replication:** All Stop Authority records are replicated in real time to at least two geographically separated HHI repository nodes. Replication lag must not exceed 5 seconds. Replication failures trigger an immediate integrity alert.

8.3 Stop Authority Invocation Integrity

Tamper Detection: Every Stop Authority invocation record carries a tamper-evident seal computed at invocation time. Continuous checksum verification (Section 5.3) applies with the highest cadence. Any seal validation failure triggers an immediate incident declaration.

Rollback Protection: Once a Stop Authority signal has been issued and its invocation record committed to the immutable audit chain, it cannot be retracted, reversed, or superseded by a single actor regardless of that actor's authorization level. Any formal revocation of a Stop Authority signal requires a documented quorum decision (minimum: three authorized Custodians), a written rationale, and a new attestation record bound to the original invocation in the audit chain.

Comprehensive Access Logging: The following events related to Stop Authority records must be logged in the immutable audit chain without exception:

- Every read access, including the identity of the reading actor and the purpose declared
- Every modification attempt, whether successful or rejected
- Every replication event and the receiving node's confirmation
- Every verification check and its outcome
- Every failed access attempt
- Every custody transfer and dual-attestation event

8.4 Stop Authority Chain of Custody

The following roles, access permissions, and quorum requirements govern the Stop Authority chain of custody:

Role	May Read	May Create	May Attest	May Revoke (Quorum)	Quorum Required
Stop Authority Custodian	Yes	Yes (dual-custody)	Yes	Yes (as quorum member)	3 of N Custodians
Metadata Integrity Officer	Yes	No	No (advisory only)	No	N/A
Audit Chain Administrator	Yes (audit read)	No	No	No	N/A
External Auditor	Yes (scoped, read-only)	No	No	No	N/A
Repository Custodian	No (unless designated)	No	No	No	N/A
Automated System / Agent	No	No	No	No	PROHIBITED

9. Cross-Repo Corroboration

9.1 Corroboration Architecture

Cross-repository corroboration is the process by which the integrity and canonical status of a metadata record held in one HHI repository is independently confirmed by comparing it against equivalent records held in other HHI repositories. Corroboration provides protection against

single-point compromise: an attacker or fault condition that corrupts a record in one repository cannot suppress detection if corroborating copies in independent repositories remain intact.

The **corroboration graph** models this architecture formally: each HHI repository is a node, and a directed edge between two nodes represents an active corroboration relationship, characterized by its tier applicability, frequency, and quorum weight. The corroboration graph must be strongly connected for all Tier 0-capable repositories — every Tier 0-holding repository must be reachable from every other via corroboration relationships.

9.2 Corroboration Protocol

The corroboration workflow follows five sequential phases:

1. **Select:** Identify the target record(s) for corroboration based on tier, time since last corroboration, and scheduled cadence.
2. **Hash:** Each participating repository independently computes the canonical hash of its copy of the record using the standard checksum protocol (Section 5.2).
3. **Compare:** The corroboration coordinator collects hashes from all participating repositories and compares them. Comparison is hash-only; no raw record content is transmitted during corroboration.
4. **Report:** A corroboration result record is generated, documenting: the target record ID, participating repositories, hashes received, comparison outcome, and timestamp. This record is written to the audit chain.
5. **Escalate:** If any discrepancy is detected, the escalation path defined in Section 9.3 is immediately initiated.

Corroboration Frequency and Quorum Requirements by Tier:

Tier	Corroboration Frequency	Minimum Participating Repositories	Quorum for Corroborated Status
Tier 0	Continuous; within 1 hour of every write	3	3 of 3 must agree (unanimous)
Tier 1	Every 4 hours	2	2 of 2 must agree (unanimous)
Tier 2	Every 12 hours	2	2 of 2 must agree

Tier 3	Every 24 hours (sampled)	1 (source + 1 replica)	Both must agree
--------	--------------------------	------------------------	-----------------

9.3 Discrepancy Classification and Remediation

Corroboration discrepancies are classified into four classes, each with a defined remediation workflow:

Discrepancy Class	Definition	Remediation Workflow	Authoritative Record Determination
Bit-Level Divergence	Identical logical record but hash mismatch due to encoding or serialization difference	Re-canonicalize all copies; re-hash; if resolved, log and close. If persists, treat as value drift.	Record with valid attestation token from original write event
Schema Drift	Record structure differs across repositories — field types, names, or shapes have changed in one copy	Schema Registry consulted; non-conforming copy quarantined; Drift Detection Engine alerted	Copy conforming to the Schema Registry version active at record creation
Missing Record	One or more repositories do not hold a copy of the expected record	Immediate alert; check replication logs; if intentional deletion, treat as integrity violation; restore from corroborated copy	Corroborated copy from remaining repositories
Conflicting Provenance	Same record ID holds different provenance bindings across repositories	All copies quarantined; full forensic audit chain trace; Metadata Integrity Officer decision; Stop	Provenance chain with earliest, cryptographically valid attestation token

		Authority notified if Tier 0	
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9.4 Corroboration Reporting

Corroboration Health Dashboard: The HHI governance tooling must provide a real-time corroboration health dashboard showing: per-tier corroboration coverage percentage, time since last corroboration for each active Tier 0 record, count of open discrepancies by class, and trend of discrepancy rate over rolling 24-hour and 7-day windows. The dashboard must be available to all authorized governance roles.

SLA Targets for Corroboration Coverage:

Tier	Coverage Target	Latency SLA (post-write)	Discrepancy Resolution SLA
Tier 0	100%	≤ 1 hour	≤ 2 hours (critical: ≤ 30 min)
Tier 1	100%	≤ 4 hours	≤ 8 hours
Tier 2	≥ 99%	≤ 12 hours	≤ 24 hours
Tier 3	≥ 95% (sampled)	≤ 24 hours	≤ 48 hours

10. Immutable Audit Chains

10.1 Audit Chain Architecture

The HHI Immutable Audit Chain is the authoritative, tamper-evident record of every consequential event in the HHI metadata lifecycle. It is designed on Merkle tree principles: each audit record includes the cryptographic hash of the preceding record, forming a chain in which

any retroactive alteration of any record — or of the chain structure itself — produces a detectable hash mismatch propagating forward through all subsequent records.

The following events are mandatorily appended to the audit chain upon occurrence:

- Every metadata write event (Tier 0 and Tier 1: all; Tier 2 and Tier 3: write events; read events only for Tier 0)
- Every checksum registration and verification result
- Every provenance binding event and orphan detection event
- Every corroboration initiation, result, and discrepancy record
- Every telemetry continuity violation and response action
- Every Stop Authority access, invocation, and custody event
- Every drift detection alert and response action
- Every schema registry modification
- Every governance epoch start and end marker

The audit chain is **append-only at the infrastructure level**. Deletion of audit records is not a permitted operation at any authorization level. Any infrastructure modification that would enable deletion of audit records is itself a critical integrity violation.

10.2 Audit Record Schema

All audit records appended to the HHI Immutable Audit Chain must conform to the following schema:

Field	Type	Required	Description
<code>auditID</code>	UUID v4	Required	Globally unique identifier for this audit record.
<code>timestamp</code>	ISO 8601 UTC	Required	Precise timestamp of the event, in UTC, to millisecond resolution.
<code>actorID</code>	URI	Required	Authenticated identifier of the actor (user, system, agent) responsible for the audited event.

actorType	Enum	Required	Classification of the actor: HUMAN, SYSTEM, AGENT, INTEGRATION.
action	Enum	Required	The audited action type: WRITE, READ, VERIFY, CORROBORATE, QUARANTINE, ATTEST, STOP_AUTHORITY, SCHEMA_CHANGE, DRIFT_ALERT, ARCHIVE, ACCESS_DENIED, and others as defined in the Schema Registry.
targetRecordID	URI	Required	Identifier of the metadata record that is the subject of the audited action.
targetTier	Enum (0-3)	Required	Metadata classification tier of the target record at the time of the event.
beforeHash	String (hex)	Conditional	Cryptographic hash of the target record's state immediately before the audited action. Required for WRITE, MODIFY, and ARCHIVE actions.
afterHash	String (hex)	Conditional	Cryptographic hash of the target record's state immediately after the audited action. Required for

			WRITE, MODIFY, and ARCHIVE actions.
<code>outcomeCode</code>	Enum	Required	Result of the audited action: SUCCESS, FAILURE, PARTIAL, REJECTED, ESCALATED.
<code>chainPreviousHash</code>	String (hex)	Required	The <code>chainCurrentHash</code> value of the immediately preceding audit record in the chain. Establishes hash linkage.
<code>chainCurrentHash</code>	String (hex)	Required	The cryptographic hash of this audit record's canonical serialization, including <code>chainPreviousHash</code> . Forms the chain link.
<code>attestationToken</code>	JWT (signed)	Required	Signed attestation token of the actor, binding their authenticated identity to this audit record at the stated timestamp.

10.3 Audit Chain Integrity Verification

Continuous Chain Validation: A background chain validation process continuously traverses the audit chain, verifying that each record's `chainCurrentHash` is correctly computed from the record body plus `chainPreviousHash`. Any hash linkage failure is immediately classified as a critical integrity violation.

Scheduled Full-Chain Replay Audits: A complete replay audit of the entire audit chain — recomputing every hash from genesis to current tip — is conducted quarterly at minimum, and

on-demand following any critical incident. Full-chain replay results are themselves recorded in a separate audit chain validation log.

Third-Party Audit Access: External auditors authorized under Section 13.2 are granted time-bounded, read-only access to audit chain segments relevant to their audit scope. Access is logged with the auditor's identity, scope, and timestamps. External auditors may independently verify chain hash linkage using their own tooling against exported chain segments.

10.4 Audit Chain Storage and Replication

- **Replication Factor:** The audit chain must be replicated to a minimum of **three geographically separated** storage nodes at all times. Replication lag must not exceed 60 seconds for Tier 0 audit records and 5 minutes for all other records.
- **Write-Once Media for Tier 0 Segments:** Audit chain segments containing Tier 0 records must be persisted to write-once or WORM (Write Once Read Many) compliant storage. This requirement applies at the infrastructure level and may not be satisfied by application-layer access controls alone.
- **Audit Chain Export Format:** Audit chain exports — produced for external audits, archival sealing, or disaster recovery — must be in JSONL format with embedded Merkle proofs for each exported segment. The export file must carry a cryptographic signature from the exporting system and a hash manifest allowing the receiving party to verify completeness and integrity of the export.

11. Runtime Drift Detection

11.1 Drift Definition and Taxonomy

Metadata drift is any unplanned, unattested deviation in a metadata field value, schema structure, relationship binding, temporal marker, or telemetry sequence from the record's last verified canonical state. Drift that has been approved through the change management process (Section 12.3) and attested via an attestation token is not drift — it is an authorized change. Only deviations without valid, contemporaneous attestation constitute drift for the purposes of this Plan.

Drift Type	Definition	Indicator Signals
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Schema Drift	A metadata field's type, name, shape, cardinality, or constraint has changed from the Schema Registry definition without a recorded schema version update	Schema validation failure; type mismatch; unexpected null or extra field
Value Drift	A metadata field value has changed from the canonical baseline without an attested write event in the audit chain	Hash mismatch on verification; baseline comparison delta without matching audit record
Relationship Drift	A provenance link, identity binding, corroboration relationship, or schema reference has changed without an attested modification event	Provenance chain break; orphan detection; corroboration graph edge missing
Temporal Drift	A timestamp field carries a value inconsistent with the governance epoch, implausible relative to adjacent events, or exhibiting clock skew beyond the permitted tolerance (± 5 seconds for Tier 0)	Timestamp out of epoch bounds; sequence anomaly; clock skew alert from time synchronization monitoring
Continuity Drift	A gap, sequence break, or silent drop in the telemetry stream as defined in Section 7	Sequence number gap; missing heartbeat; resequencing buffer overflow

11.2 Drift Detection Engine

Continuous Baseline Comparison: The Drift Detection Engine maintains a verified canonical snapshot of all active Tier 0 and Tier 1 metadata records, updated after each successfully attested write event. At defined intervals, the live state of each record is compared against its canonical snapshot. Any delta not explained by a valid, matching audit chain entry is classified as drift.

Drift Detection Interval Targets by Tier:

Tier	Detection Interval Target	Comparison Method
Tier 0	≤ 30 seconds	Continuous hash comparison against canonical snapshot + epoch root anchor
Tier 1	≤ 5 minutes	Periodic hash comparison against canonical snapshot
Tier 2	≤ 15 minutes	Periodic hash comparison (sampled for large record sets)
Tier 3	≤ 1 hour	Scheduled batch comparison

Drift Signal Pipeline: Upon detection, drift signals flow through a four-stage pipeline: (1) **Detection** — the Drift Detection Engine identifies the deviation and records a raw drift event; (2) **Scoring** — the drift is scored for severity using the formula in Section 11.3; (3) **Routing** — the scored drift signal is routed to the appropriate response queue and the relevant role's alert channel; (4) **Alerting** — human-readable alerts are dispatched to on-call contacts as specified in Section 11.5.

11.3 Drift Scoring and Severity

Drift severity is determined by a composite scoring formula incorporating four dimensions:

- **Tier Weight:** Tier 0 = 4.0, Tier 1 = 3.0, Tier 2 = 2.0, Tier 3 = 1.0
- **Field Sensitivity:** Stop Authority field = 4.0; Provenance/attestation field = 3.0; Operational field = 2.0; Descriptive field = 1.0
- **Magnitude of Change:** Complete value replacement = 3.0; Partial modification = 2.0; Structural change = 2.5; Trivial/whitespace = 0.5
- **Recency:** Within 5 minutes of last verified state = 1.5; 5–60 minutes = 1.2; Over 60 minutes = 1.0

The composite score is: `Score = Tier Weight × Field Sensitivity × Magnitude × Recency`. Score ranges map to severity levels as follows:

Score Range	Severity Level	Response Required
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0.5 – 3.0	Informational	Log and monitor; no immediate action
3.1 – 9.0	Low	Alert Repository Custodian; investigate within 24 hours
9.1 – 18.0	Medium	Alert Metadata Integrity Officer; investigate within 4 hours
18.1 – 36.0	High	Quarantine record; alert MIO and Audit Chain Administrator; investigate within 1 hour
36.1+	Critical	Quarantine; immediate Stop Authority notification; incident declaration; governance architects engaged

⚠️ AUTO-ESCALATION TO STOP AUTHORITY

Any drift event scoring at **Critical severity** automatically triggers a Stop Authority notification. This notification is non-suppressible by any automated system. The Stop Authority Custodians must acknowledge the notification within 15 minutes. Failure to acknowledge within 30 minutes triggers an escalating alert chain.

11.4 Drift Response Procedures

Containment: Upon identification of a Medium, High, or Critical drift event, the affected metadata record is immediately placed in a quarantine state. While quarantined, the record may be read for forensic purposes by authorized actors but may not be used in any governance decision, automated action, or downstream derivation. Quarantine status is logged in the audit chain.

Investigation: The investigating actor — the Metadata Integrity Officer for High and Critical events — uses the audit chain and provenance binding to trace the exact nature, origin, and timeline of the drift. The investigation must answer: What changed? When did it change? What actor or process made the change? Was there an associated attestation token that was invalid, absent, or fabricated?

Remediation: Two remediation paths are available: (1) **Canonical Restore** — if the drift is unauthorized, the record is restored from the verified canonical snapshot. The restore event is recorded in the audit chain with a full forensic note. (2) **Authorized Change Attestation** — if investigation reveals the drift was a legitimate but improperly attested change, the change may be retrospectively attested by the Metadata Integrity Officer and the responsible actor, with a documented justification. This path is available only for Tier 2 and Tier 3 records; Tier 0 and Tier 1 unauthorized drift must always follow Canonical Restore.

Post-Incident: Following closure of every High or Critical drift incident, the canonical baseline is updated with the verified clean state, detection thresholds are reviewed and revised if warranted, and a post-incident report is filed with the Governance Architects Council within 5 business days.

11.5 Drift Detection Tooling Requirements

The HHI ecosystem must maintain tooling meeting the following specifications:

- **Real-Time Streaming Comparison Engine:** Capable of comparing live metadata state against canonical snapshots at the intervals specified in Section 11.2. Must support at minimum 10,000 records per second comparison throughput for Tier 0 and Tier 1 records combined.
 - **Canonical Snapshot Registry:** An isolated, write-protected registry storing the last verified canonical snapshot of every active Tier 0 and Tier 1 record. Snapshots are updated only upon successful write-time attestation and corroboration confirmation. The snapshot registry is itself a Tier 0 asset.
 - **Alert Routing and On-Call Integration:** Drift alerts must integrate with the HHI on-call management platform, supporting: severity-based routing, acknowledgment tracking, escalation timers, and audit-logged delivery confirmation for all Critical and High severity alerts. Alert delivery failures are themselves treated as integrity events.
-

12. Governance Controls and Roles

12.1 Roles and Responsibilities

The following RACI matrix defines the responsibility assignment for the primary governance functions covered by this Plan. **R** = Responsible (does the work); **A** = Accountable (owns the outcome); **C** = Consulted (provides input); **I** = Informed (receives updates).

Function	MIO	Repo Custodian	Audit Admin	Stop Auth Custodian	Drift Operator	Corroboration Lead	Ext. Auditor
Checksum anchoring execution	A	R	I	–	I	–	–
Provenance binding enforcement	A	R	I	–	–	C	–
Telemetry continuity monitoring	A	C	I	I	R	–	–
Stop Authority record custody	I	–	I	R/A	–	–	–
Cross-repo corroboration execution	A	C	I	–	–	R	–

Audit chain administration	A	–	R	I	–	–	C
Drift detection and alerting	A	C	I	I	R	–	–
Critical drift incident response	R/A	C	C	R	R	C	–
Schema registry management	A	C	I	–	–	–	–
External audit access provisioning	A	–	R	C	–	–	I
Policy review and update	R	C	C	C	C	C	C

Legend: MIO = Metadata Integrity Officer; Repo Custodian = Repository Custodian; Audit Admin = Audit Chain Administrator; Stop Auth Custodian = Stop Authority Custodian; Ext. Auditor = External Auditor.

12.2 Access Control Requirements

Access to all HHI metadata systems is governed by role-based access control (RBAC) aligned to the roles defined in Section 12.1. The following requirements are mandatory:

- **Principle of Least Privilege:** Every actor is granted the minimum permissions necessary to fulfill their defined role responsibilities. No standing administrative access to Tier 0 metadata systems is permitted; just-in-time access provisioning is required for all Tier 0 operations.
- **Multi-Factor Authentication:** All access to Tier 0 and Tier 1 metadata systems requires multi-factor authentication. Single-factor access to these systems is a policy violation regardless of other compensating controls.
- **Privileged Access Review:** All privileged roles — Stop Authority Custodian, Metadata Integrity Officer, Audit Chain Administrator — are reviewed quarterly. Role assignments not reaffirmed within 90 days are automatically suspended pending re-authorization by the Governance Architects Council.
- **Access Revocation:** Access for any role must be revocable within 1 hour of a revocation request from the Governance Architects Council, including for geographically remote or federated system nodes.

12.3 Change Management

Standard Changes to Tier 0 and Tier 1 Metadata: All planned modifications to Tier 0 or Tier 1 metadata schemas, anchor keys, or governance epoch markers must be submitted to the Change Advisory Board (CAB) for review and approval before implementation. CAB approval must be documented with the approving members' identities, timestamps, and attestation tokens. The approved change is registered in the audit chain before implementation begins.

Emergency Change Procedure: In exigent circumstances where a change to Tier 0 or Tier 1 metadata is required before a CAB session can be convened, an emergency change may be authorized by the unanimous attestation of the Metadata Integrity Officer and at least two Stop Authority Custodians. The emergency change is implemented under active monitoring, and a full post-change provenance attestation and CAB ratification must be completed within 48 hours of implementation. Emergency changes are flagged permanently in the audit chain as

EMERGENCY_CHANGE.

13. Compliance, Audit, and Review

13.1 Internal Audit Requirements

Internal audits of all protection controls defined in this Plan are conducted on a quarterly basis by the Audit Chain Administrator, in coordination with the Metadata Integrity Officer. Each quarterly audit must assess:

- Completeness and continuity of the immutable audit chain for the audit period
- Checksum verification success rates and mismatch incident counts per tier
- Corroboration coverage and SLA compliance per tier
- Drift detection performance against tier interval targets

- Stop Authority record integrity and access log completeness
- Privileged access review completion status
- Open and resolved drift incidents and their disposition

Audit sampling methodology for Tier 2 and Tier 3 records uses stratified random sampling at a minimum 10% coverage rate, with 100% coverage mandated for all Tier 0 and Tier 1 records in every quarterly audit. Audit evidence — including raw audit chain exports, corroboration reports, and drift logs — must be preserved for 7 years.

13.2 External Audit Provisions

An annual third-party audit of the HHI Immutable Audit Chain and Stop Authority records is conducted by an independent, qualified auditing organization approved by the Governance Architects Council. The external audit must include:

- Independent hash linkage verification of the full audit chain
- Spot verification of at least 5% of Tier 0 checksum anchors against their registered epoch root values
- Review of all Stop Authority invocation records and chain-of-custody logs for the audit period
- Assessment of physical and logical controls protecting write-once and air-gapped storage systems

External auditors are granted time-bounded, read-only access to the audit chain and Stop Authority records relevant to their audit scope. Access credentials are provisioned by the Audit Chain Administrator, logged in the audit chain, and revoked within 24 hours of audit completion. External auditors must sign a confidentiality and non-disclosure agreement before access is granted. Audit findings are reported to the Governance Architects Council within 30 days of audit completion.

13.3 Policy Review Cycle

This Plan is reviewed on an annual cycle, with the next scheduled review on or before 23 May 2027. Interim out-of-cycle reviews are triggered by any of the following events:

- A major structural change to the HHI ecosystem affecting metadata flows or repository topology
- Any Stop Authority invocation event
- Any Critical-severity drift incident
- A material finding from an internal or external audit
- A change in applicable law, regulation, or industry standard materially affecting metadata governance obligations

All revisions to this Plan are subject to approval by the Governance Architects Council and are documented in the revision history table in Appendix E. No retroactive or undocumented amendments to this Plan are permitted.

13.4 Key Performance Indicators

The following KPIs are tracked continuously and reported in the governance health dashboard and quarterly audit reports:

KPI	Description	Target	Measurement Frequency	Owner
Checksum Verification Success Rate	Percentage of scheduled checksum verifications completing without mismatch	≥ 99.99%	Continuous / Daily report	MIO
Tier 0 Corroboration Coverage	Percentage of Tier 0 records corroborated within SLA following write	100% within 1 hr	Continuous / Hourly report	Corroboration Lead
Drift Detection Latency (Tier 0)	Time from drift occurrence to detection alert generation	≤ 30 seconds	Continuous / Daily report	Drift Operator
Drift Detection Latency (Tier 1)	Time from drift occurrence to detection alert generation	≤ 5 minutes	Continuous / Daily report	Drift Operator
Stop Authority Record Integrity	Percentage of Stop Authority records passing continuous checksum verification	100% at all times	Continuous	Stop Auth Custodian

Audit Chain Continuity	Number of hash linkage breaks detected in the immutable audit chain	Zero breaks	Continuous	Audit Chain Admin
Telemetry Continuity Violation Rate	Number of Major Gap or higher continuity violations per 1,000 telemetry events	< 0.01 per 1,000	Continuous / Hourly report	Drift Operator
Provenance Orphan Rate	Number of Tier 0 and Tier 1 records without valid provenance binding at any point	Zero	Continuous	MIO
Critical Drift Incident Resolution Time	Time from Critical drift alert to closure of associated incident record	≤ 4 hours	Per incident	MIO
Quarterly Internal Audit Completion	Percentage of quarterly audit deliverables completed on schedule	100%	Quarterly	Audit Chain Admin

14. Appendices

APPENDIX A – Checksum Algorithm Reference

A.1 Approved Algorithms

Algorithm	Output Size	Approved Tiers	Status	Notes
SHA-3-512	512 bits	Tier 0 (mandatory), Tier 1 (preferred)	ACTIVE	NIST FIPS 202 compliant. Preferred algorithm for all new implementations.
SHA-256	256 bits	Tier 1, Tier 2	ACTIVE	NIST FIPS 180-4 compliant. Acceptable for Tier 1; not approved for Tier 0 after 2027-01-01.
SHA-3-256	256 bits	Tier 2, Tier 3	ACTIVE	Approved for continuity integrity hashing and Tier 3 spot verification.
SHA-1	160 bits	None	DEPRECATED	Not approved for any HHI metadata integrity use. Legacy records using SHA-1 must be re-anchored within 90 days of

				this Plan's effective date.
MD5	128 bits	None	PROHIBITED	Completely prohibited in HHI metadata integrity contexts. Use constitutes a policy violation.

A.2 Algorithm Deprecation Schedule

SHA-256 deprecation for Tier 0 use: **01 January 2027**. All Tier 0 checksum anchors using SHA-256 must be migrated to SHA-3-512 before this date. Migration plans must be submitted to the Governance Architects Council by 01 October 2026.

APPENDIX B – Provenance Record Templates

B.1 JSON-LD Provenance Record Template (W3C PROV-DM Aligned)

The following template defines the canonical JSON-LD structure for an HHI provenance record. All required fields must be populated at write time. Optional fields should be populated where applicable.

```
{
  "@context": {
    "prov": "http://www.w3.org/ns/prov#",
    "hhi": "https://governance.hhi.internal/ns/prov#",
    "xsd": "http://www.w3.org/2001/XMLSchema#"
  },
  "@type": "prov:Entity",
  "prov:entity": {
    "@id": "https://repo.hhi.internal/records/{recordID}"
  },
  "prov:wasGeneratedBy": {
    "@type": "prov:Activity",
    "@id": "https://repo.hhi.internal/activities/{activityID}",
    "prov:startedAtTime": {
      "@type": "xsd:dateTime",
      "@value": "YYYY-MM-DDTHH:MM:SS.sssZ"
    }
  }
}
```

```

    },
    "prov:endedAtTime": {
      "@type": "xsd:dateTime",
      "@value": "YYYY-MM-DDTHH:MM:SS.sssZ"
    }
  },
  "prov:wasAssociatedWith": {
    "@type": "prov:Agent",
    "@id": "https://identity.hhi.internal/actors/{actorID}"
  },
  "prov:agent": {
    "@id": "https://identity.hhi.internal/actors/{actorID}"
  },
  "prov:activity": {
    "@id": "https://repo.hhi.internal/activities/{activityID}"
  },
  "prov:wasDerivedFrom": {
    "@id": "https://repo.hhi.internal/records/{sourceRecordID}"
  },
  "hhi:epochID": "{governance-epoch-uuid}",
  "hhi:attestationToken": "{signed-JWT-attestation-token}",
  "hhi:tier": "0",
  "hhi:provenanceSchemaVersion": "1.0.0"
}

```

Note: All URI templates using {placeholders} must be resolved to actual identifiers at write time. The `hhi:attestationToken` field must contain a valid, signed JWT issued by the writing actor's authenticated identity provider.

APPENDIX C – Drift Detection Configuration Reference

C.1 Reference Configuration Parameters

Parameter	Default Value	Applies To	Description
<code>drift.tier0.interval_seconds</code>	30	Tier 0	Maximum seconds between consecutive drift detection comparisons for Tier 0 records. Must not be increased above 30.

<code>drift.tier1.interval_seconds</code>	300	Tier 1	Comparison interval for Tier 1 records. Configurable in range 60–300.
<code>drift.tier2.interval_seconds</code>	900	Tier 2	Comparison interval for Tier 2 records. Configurable in range 300–900.
<code>drift.tier3.interval_seconds</code>	3600	Tier 3	Comparison interval for Tier 3 records. Configurable in range 900–3600.
<code>drift.heartbeat.tier0_interval_s</code>	10	Tier 0 telemetry	Heartbeat injection interval for Tier 0-relevant telemetry channels.
<code>drift.heartbeat.tier2_interval_s</code>	60	Tier 2 telemetry	Heartbeat injection interval for Tier 2 telemetry channels.
<code>drift.resequence.buffer_depth</code>	100	All telemetry	Maximum number of out-of-order events held in the resequencing buffer before a violation is raised.

<code>drift.resequence.timeout_seconds</code>	30	All telemetry	Maximum time an event may remain in the resequencing buffer before a violation is raised.
<code>drift.alert.critical_ack_timeout_s</code>	900	Critical alerts	Seconds before an unacknowledged Critical drift alert triggers escalating alert chain. Must not exceed 900 (15 min).
<code>drift.snapshot.update_on_attest</code>	true	All tiers	When true, canonical snapshot is updated only upon successful write-time attestation and corroboration. Must not be set to false for Tier 0.

i Configuration Change Control

Changes to any drift detection configuration parameters for Tier 0 or Tier 1 are subject to CAB approval (Section 12.3). Configuration changes are logged in the audit chain as `SCHEMA_CHANGE` events.

APPENDIX D – Corroboration Quorum Rules

D.1 Formal Quorum Definitions

A metadata record achieves **Corroborated** status when the following quorum conditions are satisfied within the specified SLA window following the most recent write event:

Tier	Minimum Repositories (N)	Agreement Required (Q)	Quorum Formula	SLA Window
Tier 0	3	3 (unanimous)	Q = N (100%)	60 minutes
Tier 1	2	2 (unanimous)	Q = N (100%)	4 hours
Tier 2	2	2 (unanimous)	Q = N (100%)	12 hours
Tier 3	2 (source + 1)	2	Q = N	24 hours

D.2 Quorum Failure Handling

If quorum cannot be achieved within the SLA window due to repository unavailability (not discrepancy):

- **Tier 0:** Record is placed in `PENDING_CORROBORATION` state immediately. Stop Authority Custodian notified. Record may not be used for governance decisions until corroboration is achieved or a formal exception is granted by the Governance Architects Council.
- **Tier 1:** Alert to Metadata Integrity Officer. Record flagged as `PENDING_CORROBORATION`. 4-hour grace extension permitted once; after which treated as Tier 0 failure handling.
- **Tier 2 and Tier 3:** Alert logged; repository availability investigated; corroboration attempted at next cycle.

APPENDIX E – Document Control and Revision History

E.1 Revision History

Version	Date	Author / Role	Status	Summary of Changes
1.0.0	23 May 2026	Metadata Integrity Officer, HHI SIO	ACTIVE	Initial release. Full document covering all seven protection pillars. Approved by HHI Governance Architects Council.
(future)	—	—	—	Reserved for subsequent revisions. All revisions to be recorded here with full author, date, approver, and change summary.

E.2 Document Approval Record

Role	Name / Identifier	Action	Date	Attestation Reference
Primary Author	Metadata Integrity Officer, HHI SIO	Drafted	23 May 2026	ATT-MDP-001-D RAFT-001

Governance Review	HHI Governance Architects Council	Reviewed & Approved	23 May 2026	ATT-MDP-001-G AC-APPROVE-001
Stop Authority Custodian (1)	SAC-ROLE-001	Attested (Tier 0 sections)	23 May 2026	ATT-MDP-001-S AC1-001
Stop Authority Custodian (2)	SAC-ROLE-002	Attested (Tier 0 sections, dual-custody)	23 May 2026	ATT-MDP-001-S AC2-001
Audit Chain Administrator	ACA-ROLE-001	Registered in Audit Chain	23 May 2026	AUDIT-CHAIN-RECORD-MDP-001

HHI-GOV-MDP-001 | Version 1.0.0 | Effective 23 May 2026

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